1. Clinical Question & Core Hypothesis

Clinical Question

What molecular mechanisms determine whether immune structures translate into clinical benefit?

Working Hypothesis

Primary hypothesis: In bladder cancer, a high stromal-to-tumor ratio (TSR) driven by TGFB1-mediated fibrosis creates an immune-excluding tumor microenvironment that preferentially supports survival, clonal expansion, and invasion of MSH6-deficient, genomically unstable tumor cells. The combination of high TSR with TGFB1 upregulation and MSH6 deficiency predicts shorter progression-free intervals (PFI) compared with tumors lacking this triad. Secondary modifiers such as circulating tumor cells (CTCs), microsatellite instability (MSI), and GREM1 signaling may modulate the magnitude of the effect but are not required anchors. Mechanistically, TGFB1-induced stromal remodeling reduces cytotoxic T-cell infiltration and promotes ECM stiffening, enabling immune evasion and enhanced invasiveness of MSH6-deficient clones, driving earlier progression. Testable predictions include: (1) a significant interaction between TSR, TGFB1 status, and MSH6 deficiency for PFI; (2) TGFB1 pathway inhibition or stromal normalization will restore immune infiltration and slow progression in model systems; (3) higher CTC burden will correlate with high TSR/MSH6-deficient tumors, supporting a mediator role for CTCs but not as core anchors.

Element	Value
Imaging feature	Stromal-to-tumor ratio (TSR)
Anchor gene(s)	MSH6, TGFB1
Primary endpoint	PFI

2. Why This Gene — Not Others

Primary hypothesis gene(s)

MSH6: MSH6-driven genomic instability translate into early progression? 2) Tier 1 paths used: I built on the MSH6 prognostic path (links genomic instability to outcome) and the TGF-β fibrogenic path (links stromal remodeling / immune exclusion to invasive behavior); I retained the CTC/liquid-biopsy path as a mechanistic readout but not as a core anchor to keep the hypothesis sharp.

TGFB1: Selected as a core mechanistic driver from Tier 1 knowledge graph analysis.

Alternative / secondary genes considered

Tier	Gene	Role / Reason excluded from core
Secondary	CTCs	Supporting modifier from knowledge graph

Secondary	MSI	Supporting modifier from knowledge graph
Secondary	GREM1	Supporting modifier from knowledge graph

3. Clinically Testable Predictions

Prediction 1: retrospective clinical analysis correlating TSR (histology/radiomics) and TGFB1/MSH6 status with PFI, including interaction terms;

Prediction 2: in vitro organoid/3D co-cultures of bladder cancer cells with MSH6 deficiency and cancer-associated fibroblasts under TGFB1 treatment to assess proliferation, invasion, ECM remodeling, and immune cell killing; test TGFB1 inhibitors (e.g., galunisertib) for reversal;

Prediction 3: in vivo mouse models with MSH6-deficient tumors and TGFB1■driven stroma to measure TSR, immune infiltration (CD8+ T cells), tumor growth, and metastasis; evaluate TGFB1 blockade and stromal normalization as therapeutic interventions. This framework yields clear, testable endpoints focused on mechanism-driven progression.

4. Minimum Viable Validation Plan

A. Retrospective Cohort Analysis

Data source: TCGA-BLCA (Firehose Legacy), ~400 patients - Cox PH model with interaction term for the imaging feature and anchor gene - Kaplan-Meier 2x2 stratification

B. Tissue-Level Validation

Methods: Multiplex IF / IHC / spatial transcriptomics

Key readouts: - Immune structure markers (e.g., CD20, CD3, PNAd) - Effector cell localization (CD8) relative to tumor nests - Anchor gene protein / phospho-protein expression - Stromal features (collagen density by SHG, if applicable)

Goal: Confirm that the proposed mechanism operates at the spatial / tissue level.

C. Functional Validation

Models: Ex vivo organotypic cultures, 3D co-culture, syngeneic mouse models

Perturbations: Genetic (KO/rescue) or pharmacologic modulation of anchor gene

Readouts: T-cell infiltration, cytotoxicity (granzyme B / IFN-gamma), tumor growth / survival

Goal: Establish causal relationship between anchor gene activity and immune efficacy.

5. Strengths, Caveats & Alternative Directions

Strengths

- Interaction-first biomarker strategy — asks WHY the imaging feature sometimes fails, not just whether it predicts outcome.

Caveats & Risks

- Hypothesis is computationally derived and requires experimental validation.

Alternative Directions

- CTCs: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.
- MSI: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.